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Embedded capacity-computer module for microservice load balancing and distribution

Abstract

Disclosed herein are system, method, and computer program product embodiments for microservice load balancing and distribution using an embedded computer capacity module. An embodiment operates by retrieving an application programming interface (API) request from a client. The embodiment stores objects representing the API request in a job detail database containing details related to the API request. The embodiment determines an available bandwidth of the service instance. The embodiment transmits the determined available bandwidth of the service instance to a job processor. The embodiment selects tasks from the objects representing the API request stored in the job detail database based on the determined available bandwidth of the service instance. The embodiment executes the selected tasks.

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Background/Summary

BACKGROUND

(1) With the advent of cloud-native applications, new cloud-native environments have been designed for automating mass computer application, deployment, and management. In these cloud-native environments, solutions for computer application deployment commonly involves running numerous microservices. For example, cloud-native data processing systems may run multiple instances of each microservice and application. As a result, numerous service instances must share the load of the tasks the service instances are assigned to execute.

(2) Legacy cloud-native data processing systems often process load balancing through an external load balancer. However, these legacy systems have no knowledge of whether particular service instances are actually capable of handling a task for execution at a given time. Microservices generally have no control over external load balancers. As a result, microservices may be unable to transmit a signal to the external load balancer that the microservice instance cannot process particular tasks.

(3) Consequently, due to the disproportionate allocation of microservice loads, service instances may be overwhelmed and fail to process assigned tasks. To properly balance and distribute microservice loads, a self-balancing microservice load balancing and distribution system is necessary to prevent overwhelming particular service instances.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are incorporated herein and form a part of the specification.
- (2) FIG. 1 is a block diagram illustrating a cloud-native data processing system, according to some embodiments.
- (3) FIG. 2A is a block diagram illustrating an example of cloud routing within a cloud-native data processing system, according to some embodiments.
- (4) FIG. 2B is a block diagram illustrating an example of cloud routing within a cloud-native data processing system using an external service load balancer, according to some embodiments.
- (5) FIG. 2C is a block diagram illustrating an example of cloud routing within a cloud-native data processing system using an external service load balancer for each microservice, according to some embodiments.
- (6) FIG. 3 is a block diagram illustrating an example of an embedded capacity-computer module for microservice load balancing and distribution, according to some embodiments.
- (7) FIGS. 4A-4D are an example of a job detail database, according to some embodiments.
- (8) FIG. 5 is a flowchart illustrating a method for microservice load balancing and distribution through an embedded capacity-computer module, according to some embodiments.
- (9) FIG. 6 is an example computer system useful for implementing various embodiments.
- (10) In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

- (11) Provided herein are system, apparatus, device, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for microservice load balancing and distribution through an embedded capacity-computer module. A load balancer plays an important role in distributing and balancing the load for microservices equitably in a cloud-native ecosystem. In legacy cloud-native environments, data processing systems run several instances of a microservice to process incoming API requests. These service instances need to share the load of the jobs the service instances are designed to execute. As a result, legacy load balancers distribute tasks agnostically among microservice instances without any knowledge of each microservice instance's bandwidth.
- (12) This approach is not optimal because an external load balancer may overload microservice instances with a full bandwidth while overlooking a microservice instance that may have free bandwidth to process tasks for incoming API requests. External load balancers allocating disproportionate loads amongst microservices can lead to costly and draconian job failures. Therefore, a technological solution is needed to create an ecosystem in which each microservice instance can cooperatively distribute and balance the load of incoming API requests in an equitable fashion. This technological solution would allow each microservice instance to self-regulate load distribution by computing parameters for available capacity unique to that microservice instance without an agnostic external load balancer.
- (13) FIG. 1 is a block diagram illustrating a cloud-native data processing system, according to some embodiments. The system **100** is an example embodiment of a cloud-native data processing system. The system **100** may contain one or more applications **104** (**104-A**, **104-B**, **104-C**, . . . , **104-X**, . . .). Each application **104** may be a distributed application comprising microservices **110** (**110-A1**, **110-A2**, **110-A3**, . . . , **110-X1** . . .). Each microservice **110** may have several service instances **108** (**108-A1**, **108-A2**, **108-A3**, . . . , **108-X1(1)**, **108X1(2)**, . . .). The applications **104** may be cluster managed by a cluster orchestrator. The cluster **102** may simultaneously host and run multiple applications **104**. The set of applications **104** in a cluster **102** can be dynamic, in which the composition of each application **104** may change over time (e.g., due to upgrades) and the set of applications **104** themselves may also change over time.

(14) In order to store the underlying data in a cloud orchestrator, the system **100** can use a per-application repository **106** (**106-A**, **106-B**, **106-C**, . . . **106-X**, . . .). An entire application **102** can be used as the unit while storing application data in a repository **106**. The system **100** can store the data from all the service instances **108** in a single repository **106**, such as an object store bucket. Data in a service instance **108** may all belong to the same application **104** and data from different applications **104** may be stored in separate repositories **106**. The data may be located in separate service instances **108** each with their own access restrictions.

(15) As discussed above, legacy load balancers do not have knowledge of the loads of specific service instances **108** in a cluster **102** and thereby may disproportionately allocate microservice loads. Therefore, service instances **108** may be overwhelmed and fail to process assigned tasks. To illustrate this technical problem and describe the features and functions of system **100**, reference is made to two illustrative example microservices. This example illustrates employing a cloud-native data processing system **100** to execute several service instances **108** for two microservices **110** in a cluster **102**.

(16) In this example, cloud-native data processing system **100** contains a first microservice **110** for exporting stories (referred to as “microservice **110-A1**” for illustration purposes) and a second microservice **110** for delivering messages (referred to as “microservice **110-A2**” for illustration purposes). The microservice **110-A1** for exporting stories is responsible for running a headless browser, which may generate a story or report of analytics derived within a particular application **104**. Microservice **110-A1** may then export the story to a pdf or pptx file. As a result, this microservice **110-A1** optimizes its resources by parallelizing multiple export jobs. For example, microservice **110-A1** may be handling up to four exports at once.

(17) While microservice **110-A1** may parallelize multiple export jobs, microservice **110-A1** may have to parallelize export jobs for stories of different sizes. For example, a story may be a simple single page story, which microservice **110-A1** may open and export in less than ten seconds. On the other hand, microservice **110-A1** may need to open and export heavier stories, which may take more than thirty minutes for microservice **110-A1** to complete the task. This may have a direct bearing on memory and disk, as the size of the story exported may be too large.

(18) On the other hand, microservice **110-A2** for delivering messages may be responsible for transmitting e-mails from a particular application **104** in cluster **102**. Microservice **110-A2** may connect to different servers using Simple Mail Transfer Protocol and send e-mails to designated recipients. Similar to microservice **110-A1**, microservice **110-A1** may need to deliver e-mails that contain attachment of varying sizes. Moreover, if there are multiple recipients of an e-mail, microservice **110-A2** may need to split the e-mail into independent e-mails to each recipient. Therefore, the number of recipients and the size of the attachments affects the load of the assigned task.

(19) As a result, the service instances **108** in microservice **110-A** and microservice **110-B** cannot complete the tasks within comparable times and/or with comparable resources. The service instances **108** will need a mechanism to reject certain jobs assigned for execution until the service instances **108** have freed up the bandwidth to execute further tasks. However, legacy external load balancers only forward API requests and are not aware of the internal load situation of the service instances **108** for microservice **110-A** and microservice **110-B**. Thereby, legacy external load balancers simply forward the request based on its own implemented algorithm, such as round robin.

(20) FIG. 2A is a block diagram illustrating an example of cloud routing within a cloud-native data processing system, according to some embodiments. FIG. 2 is described with reference to FIG. 1. Cloud-native data processing system **100** may retrieve API requests **205** from one or more API clients **202**. To process API request **205**, cloud-native data processing system **100** may use an application router **210** to route any incoming API request **205** to the relevant service instances **108**. Legacy cloud-native data processing systems generally use a central application router **210** for an

entire application **104** to route API request **205** to the relevant service instances **108**.

(21) As a result, application router **210** may be an infrastructure service for cloud-native data processing system **100** that cannot be controlled and/or customized for individual service instances **108** used to process API request **205**. Application router **210** may use routing techniques known to a person of ordinary skill in the art to route API request **205** to the relevant service instances **108**. In legacy cloud-native data processing systems, application router **210** uniformly routes API request **205** to the relevant microservices **110** and service instances **108**. For example, application router **210** may use a technique such as round robin to the load of API request **205** among the relevant service instances **108**. By using the round robin technique to route API request **205**, application router **210** is entirely agnostic to the current bandwidth of the service instances **108**.

(22) Therefore, because application router **210** may be a centralized router for an entire application **104**, a service developer that is developing only the service **110** and/or the service instances **108** may not have any control over the load balancing or distribution of API request **205** to particular services **110** or service instances **108**. However, oftentimes services **110** and service instances **108** may need special or unique load distribution capabilities. To address this problem, cloud-native data processing system **100** may use additional service load balancer **215** for each particular service type, as described in FIG. 2B. A service load balancer **215** would need to determine the particular service instances **108** that can be assigned a task for API request **205** based on some availability metrics that the service instances **108** may transmit to the service load balancer **215**.

(23) For purposes of illustration, service instances **108** (**108-A1**, **108-A2**, and **108-A3**) are instances of microservice **110-A1** for exporting stories, as described above. In this implementation, service load balancer **215** may distribute load only amongst service instances **108** of microservice **110-A1** for exporting stories. While this may be technically feasible, service developers will need to write two separate microservices **110**: one microservice for the actual service for exporting stories and one microservice for the service load balancer **215**. As a result, enterprises will need to expend significant overhead to separately maintain the life cycle of service load balancer **215**. As a result, cloud-native data processing systems will need to free up additional disk space for the service load balancer **215**. Otherwise, enterprises may face catastrophic consequences in the case service load balancer **215** fails or becomes unavailable. Therefore, cloud-native data processing system **100** will need to run multiple instances of service load balancer **215**.

(24) Even with the implementation in FIG. 2B, the solution for microservice load balancing and distribution is still incomplete. For example, as shown in FIG. 2C, service load balancer **215-A1** and service load balancer **215-A1** may both simultaneously detect that service instance **108-A1** has spare bandwidth. Both service load balancer **215-A1** and service load balancer **215-A2** may accordingly assign separate tasks to service instance **108-A1**. This may cause issues because the separate tasks service load balancer **215-A1** and service load balancer **215-A2** assigned may exceed the spare bandwidth of service instance **108-A1**. Therefore, the service load balancers **215** would need further implementation to lock particular service instances **108** and assign a task.

(25) Considering the significant overhead costs for further implementation (as illustrated in FIGS. 2A and 2B) and existing architectural models for load balancing and distribution, legacy cloud-native data processing systems generally use load balancers external to the microservice **110** and service instances **108** implementations. Therefore, a solution is needed to optimize microservice load balancing and distribution by enabling service instances **108** to intelligently determine which tasks the service instances **108** would need to process for API request **205** without an external load balancer.

(26) FIG. 3 is a block diagram illustrates an example of an embedded capacity-computer module for microservice load balancing and distribution, according to some embodiments. FIG. 3 is described with reference to FIG. 1 and FIG. 2. As discussed above, a solution is needed to distribute and balance loads for each service instance **108** without an external load balancer. An embedded capacity-computer module for each service instance **108** enables load balancing and

distribution without the consequences and overhead costs of an external load balancer. Each microservice **110** for applications **104** may include a job detail database **325**. Each service instance **108** for application **104** may include, but is not limited to, API layer **310**, capacity computer module **315**, and job processor **320**.

(27) In order to distribute and balance the load of service instances **108**, system **100** may not immediately process a job when a service instance is assigned an API request **205**. Instead, system **100** defers the job processing until a service instance **108** can process API request **205**. To defer API request **205**, service instance **108** transmits an incoming API request **205** into job detail database **325** as an object. Job detail database **325** may be a data structure that stores API requests **205** in a job queue. Job detail database **325** may contain job details for each incoming API request **205** including, but not limited to, the time of arrival of API request **205**, the time of insertion of the object representing API request **205** in job detail database **325**, and other relevant information to the API request **205**. Each incoming API request **205** may be stored into a common queue in job detail database **325**.

(28) Job detail database **325** may be a data store. Job detail database **325** may be a relational database, a NoSQL database or other horizontally scaling database, or any other database adhering to a suitable database design methodology. In an embodiment, the data store may implement a centralized storage area network (SAN), network-attached storage (NAS), redundant array of independent disks, and/or any other configuration of storage devices to supply sufficient storage capacity to store database tables and supporting structures. Sufficient storage may alternatively exist in any other physically attached magnetic storage, cloud storage, or additional storage medium. In an embodiment, the data store may deploy a hard-disk interface, such as ATA, SATA, SCSI, SAS, and/or fibre for interfacing with storage mediums.

(29) Each service instance **108** may have an internal, embedded capacity computer module **315**. The capacity computer module **315** for each service instance **108** may continuously monitor the capacity and/or available bandwidth for the service instance **108** embedding its respective capacity computer module **315**. Capacity computer module **315** may continuously, or at frequent intervals, monitor and/or compute the available bandwidth of its respective instance **108**. Capacity computer module **315** may consider various parameters and metrics to compute capacity and/or available bandwidth for each service instance **108** including, but not limited to, available memory, available disk space, the amount of memory required to process certain API requests **205**, etc. Capacity computer module **315** may communicate to job processor **320** the capacity and/or available bandwidth service instance **108** has for a designated number of jobs.

(30) Each service instance **108** may have a job processor **320**. Job processor **320** selects API requests **205** from job detail database **325** to process up to the designated number of jobs capacity computer module **315** communicated to job processor **320**. Service instances **108** may then lock the objects of the API requests **325** in job detail database **325** after job processor **320** selected the API requests **205** to process from job detail database **325**. As a result, no more than one service instance **108** may select and/or process the same API request **205** stored in job detail database **325**. Service instances **108** may use any algorithm known to a person of ordinary skill in the art to lock API requests **205** in job detail database **325**.

(31) In some embodiments, microservices **110** may have templates for service development. These templates may define standards for service development using interface definitions. For example, microservice **110** may include a functionality to include an interface for computing the capacity of each service instance **108** (i.e., `computeCapacity() { }`). In some embodiments, the microservice **110** itself may designate itself as a self-balancing service. Self-balancing microservices **110** may override an interface method (e.g., `computeCapacity() { }`) according to the needs of that microservice **110**.

(32) To describe the solution architecture and functions of system **300**, reference is made to the illustrative example microservices for exporting stories, as described above in FIGS. 1-2. System

100 may receive several API requests **205** from API client **202** to export stories. System **100** may route API requests **205** to the application router **210** for the relevant application **104**, and in this instance, application router **210-A** for application **104-A**. Application **210-A** may then route the particular tasks of API requests **205** to the relevant service instance **108**. In this case, application router **210-A** routes API requests **205** to API layer **310-A** of service instance **108-A(1)** for exporting stories. Before processing API requests **205**, service instance **108-A(1)** transmits API requests **205** and job detail database **325-A** stores objects of the incoming API requests **205**.

(33) Capacity computer module **315-A(1)** then computes capacity and/or available bandwidth for service instance **108-A(1)**. The algorithm may include various parameters and metrics to determine whether service instance **108-A(1)** has capacity and/or available bandwidth for processing API requests **205** to export stories. In this case, capacity computer module **315-A(1)** may consider the following metrics. First, capacity computer module **315-A(1)** may consider available memory. Memory may be considered to be a crucial factor in an entire workflow for starting up an instance of a headless browser, loading a story, and then exporting the story to a PDF file. As a secondary consideration, capacity computer module **215-A(1)** may consider available disk space, which may become relevant if the generated artifacts are saved on a disk.

(34) Finally, capacity computer module **315-A(1)** may consider metrics that determine the nature of the story included in API requests **205-A**. For example, capacity computer module **315-A** may consider how much memory is required to load and export stories to PDF or PPTx files. These metrics may be a classifier metrics that classifies a story as a small, medium, or large files. Depending on the amount of memory available at that time, capacity computer module **315-A(1)** may use the resulting computation to inform its determination of how many tasks job processor **320** can process. These metrics are only highlighted for purposes of illustration. However, a person of ordinary skill in the art may consider any relevant parameters, metrics, and/or algorithms a capacity computer module **315** may incorporate into its computation to determine the capacity and/or available bandwidth for service instances **108**.

(35) As discussed above, microservice **110-A1** for exporting stories is responsible for running a headless browser, which may generate a story or report of analytics derived within a particular application **104**. Microservice **110-A1** may then export the story to a pdf or pptx file. As a result, this microservice **110-A1** optimizes its resources by parallelizing multiple export jobs.

Microservice **110-A1** may run several headless browser. The number of browser instances that can run may determine on the concurrency of export jobs service instance **108** can handle. For example, microservice **110-A1** may only be able to handle up to four exports at once.

(36) After capacity computer module **315-A(1)** determines the capacity and/or available bandwidth for service instance **108-A(1)** to process particular jobs for exporting stories stored in job detail database **325-A**, capacity computer module **315-A(1)** communicates the designated number of tasks from API requests **205** job processor **320** can process. Therefore, job processor **320** may select a job from job detail database **325-A** for running a headless browser, generating a story of analytics, exporting the story to a pdf or pptx file, etc. However, job processor **320** may only select these jobs up to the designated number of tasks from API requests **205** job processor **320** may process. After job processor **320** selects jobs to process from job detail database **325-A**, service instance **108-A(1)** may lock these jobs in job detail database **325-A**. As a result, only service instance **108-A(1)** may select and process the relevant jobs to process API requests **205**. Job processor **320** may then process the jobs for exporting stories selected from job detail database **325-A**.

(37) As a result, an embedded computer capacity module **315** enables service instances **108** to determine its own capacity and available/bandwidth without the costs and consequences of an external load balancer. As discussed previously, legacy load balancers could not communicate with specific service instances to determine capacity and/or available bandwidth. By embedding computer capacity module **325** for each service instance **108**, the service instances **108** in each application **104** cooperate as good citizens to distribute and balance the load of API requests **205** by

self-regulating its available capacity. Each service instance **108** may continuously, or at frequent intervals, monitors and computes its capacity and/or available bandwidth, which prevents specific service instances **108** from being idle or other service instances **108** from being overloaded. As each service instance **108** for applications **104** are proactive and involved citizens, the entire load of API requests **205** is well distributed, managed, and balanced among the service instances **108**.

(38) FIG. **4** is an example embodiment of job detail database **325**, according to some embodiments. FIG. **4** is described with reference to FIGS. **1-3**. As discussed above, job detail database **325** may be a data structure that stores API requests **205** in a job queue. Job detail database **325** may contain job details for each incoming API request **205** and other relevant information to the API request **205**. To describe the job detail database **325**, reference is made to the illustrative example microservices for exporting stories, as described above in FIGS. **1-3**.

(39) Referring to FIG. **4A**, as discussed in FIG. **3**, before processing API requests **205**, microservice **110-A(1)** transmits API requests **205** and job detail database **325-A** stores objects of the incoming API requests **205**. The API requests **205** may be stored in a table structure as illustrated in FIG. **4A**. In this example embodiment, “TABLE export_job” is a table that stores job details for an incoming API request **205**, such, as the timestamp the job detail database received the API request **205**.

(40) The job details may also include variables, such as, “TASK_ID VARCHAR(256),” which is an ID generated for the particular API request **205**, and “TIME STAMP TIMESTAMP,” which may represent the timestamp that the API request was inserted and entered into the queue. Job details may also include variables such as “NUM_RETRIES INT4,” which represents a counter of how many times an API request **205** failed to be processed because an API request **205** may remain in a table or queue until the request is completed successfully or enters into a failed state.

(41) As this is a multi-tenant solution in a cloud environment, job detail database **325** may also include details or variables related to the tenants, such as, “TENANT VARCHAR(256),” which is a unique identifier that represents which tenant an API request **205** belongs to.

(42) Finally, in this example embodiment, job detail database may also include objects representing the details of API request **205**, such as, “EXPORT_OPTIONS jsonb,” which represents the details of the API request **205** itself. This may include details similar to the payload. For example, using the microservices example for exporting stories, microservice **110** may receive the following API request **205**, as follows:

(43) TABLE-US-00001 { “id”: “{{id}}”, “tenant”: “{{tenant}}”, “tenantName”: “{{tenantName}}”, “format”: “PDF”, //the exported artifact can either be PDF or PPT “url”: “{{url}}”, //The story URL which has to be exported. “bookmarks”: [//The story can have multiple bookmarks which are chosen to //exporting. Bookmarks is set of filters. { “bookmarkId”: “STORY”, “internalId”: “{{Sguld}}”, “includeAppendix”: true, “pages”: [], “pageLayout”: { “rows”: “ALL”, “columns”: “ALL” } // Some details for the actual exporting. }] }

(44) This payload may then be stored as “EXPORT_OPTIONS jsonb” in job detail database **325**, as illustrated in FIG. **4A**.

(45) Referring to FIG. **4B**, job detail database **325** may also include a table structure that tracks executing API requests **205**. In this example embodiment, “TABLE executing_job” is a table that keeps records of executing API requests **205**. Once a microservice **110** is assigned an API request **205** and processes the API request **205**, an entry is inserted into this table. This table may include identification variables for tasks and tenants, as described in FIG. **4A**, such as, “TASK_ID VARCHAR(256)” and “TENANT VARCHAR(256).” Additionally, to track the executing jobs, this table may store identification variables, such as, “INSTANCE_ID VARCHAR(256),” which is an ID for the microservice **110** executing the API request **205**, and “INSTANCE_INDEX INT4,” which is an index of microservice **110**.

(46) Once an API request **205** is assigned to a microservice **110**, it may take some time before the

actual API request **205** is executed. For this reason, it may be valuable to incorporate metrics related to the time it takes to execute API requests **205**. Accordingly, this table may include variables and metrics to track executing API requests **205**, such as, “START_TIMESTAMP TIMESTAMP,” which captures a specific time when an export job starts for the microservice **110-A(1)** for exporting stories.

(47) As discussed above, after job processor **320** selects jobs to process from job detail database **325-A**, microservice **110-A(1)** may lock these jobs in job detail database **325-A**. Referring to FIG. **4C**, job detail database **325-A** may include a table structure for locking the job detail-database. In this example embodiment, “TABLE export_job_lock” is a table that locks job detail database **325** to ensure the same API request **205** does not get picked up by multiple microservices **110**. Accordingly, microservice **110** may acquire a lock and select jobs from the “export_job” table illustrated in FIG. **4A**.

(48) This table may include boolean variables, such as, “LOCKED BOOLEAN,” which indicates whether a particular API request **205** is locked. A value of true would indicate that API request **205** is locked and a value of false would indicate that API request **205** is currently unlocked. The table may also track the microservice **110** an API request **205** was assigned to by storing variables, such as, “LOCKED_BY VARCHAR (256),” which is an identification variable for the microservice **110** that locked a particular API request **205**. This table may also include variables and metrics relevant to tracking locked API requests **205**, such as, “LOCK_GRANTED_AT TIMESTAMP,” which is a variable that represents the time at which a particular API request **205** was locked.

(49) Job detail database **325** may also include a table structure that stores runtime details for API requests **205**. In this example embodiment, “TABLE artifact gen detail” is a table that contains runtime details about microservice **110-A(1)** for exporting stories for a particular API request **205** when and after microservice **110-A(1)** has exported a story. The table may include details about the task by storing identification variables and the tenant such as “TASK_ID VARCHAR(256)” and “TENANT VARCHAR(256).”

(50) The table may include variables representing any errors at runtime, such as, “ERROR VARCHAR(256),” which is a variable that represents whether an error occurred when microservice **110-A(1)** exported any stories, and “ERROR_DETAIL VARCHAR(1024),” which may represent full error details.

(51) The table may include variables unique to microservice **110-A(1)** for exporting stories. For example, the table may include a variable, such as, “CHECKSUM VARCHAR(1024),” which can check whether there was any corruption in a story when an artifact was uploaded or sent as an e-mail, “LOGS jsonb,” which can generate important logs for stories that were exported as part of API request **205**, “BASE64STRING bytea,” which is the final story to be sent to recipients, and “UPLOAD_TIME TIMESTAMP,” which is a variable that represents the time of uploading a story for further delivery.

(52) Accordingly, according to this example embodiment, job detail database **325** can store job details related to an API request **205**, track executing API requests **205**, lock API requests **205**, and store runtime details related to API requests **205**. This example is only provided for illustrative purposes. Job detail database **325** may store any information in any format known to a person of ordinary skill in the art for microservice load balancing and distribution.

(53) FIG. **5** is a flowchart illustrating a method for microservice load balancing and distribution using an embedded computer capacity module, according to some embodiments. FIG. **5** is described with reference to FIGS. **1-4**. Method **500** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all steps may be needed to perform the disclosure provided herein. Further, some of the steps may be performed simultaneously, or in a different order than shown in FIG. **8**, as will be understood by a person of ordinary skill in the art.

(54) At **502**, service instances **108** retrieves API requests **205**. System **100** receives API requests **205** from API client **202**. System **100** routes API requests **205** to the application router **210** for the relevant application **104**. Application **210-A** then routes the particular tasks of API requests **205** to the API layer **310** of the relevant service instance **108**. Application router **210** determines the relevant service instance **108** based on the nature of the API requests **205**.

(55) At **504**, service instances **108** stores API requests **205** as objects in job detail database **325**. In order to distribute and balance the load of service instances **108**, system **100** does not immediately process a job when a service instance is assigned an API request **205**. Instead, system **100** defers the job processing until a service instance **108** can process API request **205**. To defer API request **205**, service instance **108** stores an incoming API request **205** into job detail database **325** as an object. Job detail database **325** may be a data structure that stores API requests **205** in a job queue. Job detail database **325** may contain job details for each incoming API request **205** including, but not limited to, the time of arrival of API request **205**, the time of insertion of the object representing API request **205** in job detail database **325**, and other relevant information to the API request **205**. Each incoming API request **205** is stored into a common queue in job detail database **325**.

(56) At **506**, service instances **108** computes capacity and/or available bandwidth for service instances **108**. Each service instance **108** may have an internal, embedded capacity computer module **315**. The capacity computer module **315** for each service instance **108** may continuously monitor the capacity and/or available bandwidth for the service instance **108** embedding its respective capacity computer module **315**. Capacity computer module **315** may continuously, or at frequent intervals, monitor and/or compute the available bandwidth of its respective instance **108**. Capacity computer module **315** may consider various parameters and metrics to compute capacity and/or available bandwidth for each service instance **108** including, but not limited to, available memory, available disk space, the amount of memory required to process certain API requests **205**, etc.

(57) At **508**, service instances **108** transmits the capacity and/or available bandwidth for service instances **108**. Capacity computer module **315** may communicate to job processor **320** the capacity and/or available bandwidth service instance **108** has for a designated number of jobs.

(58) At **510**, service instances **108** selects tasks from job detail database based on the capacity and/or available bandwidth for service instances **108**. Each service instance **108** may have a job processor **320**. Job processor **320** selects API requests **205** from job detail database **325** to process up to the designated number of jobs capacity computer module **315** communicated to job processor **320**. In some embodiments, microservices **110** may have templates for service development. These templates may define standards for service development using interface definitions. For example, microservice **110** may include a functionality to include an interface for computing the capacity of each service instance **108** (i.e., `computeCapacity(){}`). In some embodiments, the microservice **110** itself may designate itself as a self-balancing service. Self-balancing microservices **110** may override an interface method (e.g., `computeCapacity(){}`) according to the needs of that microservice **110**.

(59) At **512**, service instances **108** locks job detail database **325**. Service instances **108** may then lock the objects of the API requests **205** in job detail database **325** after job processor **320** selected the API requests **205** to process from job detail database **325**. As a result, no more than one service instance **108** may select and/or process the same API request **205** stored in job detail database **325**. Service instances **108** may use any algorithm known to a person of ordinary skill in the art to lock API requests **205** in job detail database **325**.

(60) At **514**, service instances **108** executes the jobs selected from job detail database **325**. Service instances **108** may execute jobs selected from job detail database **325** for API requests **205** retrieved by a different service instance **108**.

(61) Various embodiments can be implemented, for example, using one or more computer systems, such as computer system **600** shown in FIG. 5. FIG. 5 is described with reference to FIGS. 1-5.

Computer system **600** can be used, for example, to implement method **500** of FIG. 5. For example, computer system **600** can implement and execute a set of instructions comprising retrieving API requests **205**, transmitting API requests **205** to job detail database **325**, computing the capacity and/or available bandwidth for service instances **108**, transmitting the capacity and/or available bandwidth of service instances **108** to job processor **320**, selecting tasks from job detail database **325**, locking job detail database **325**, and/or executing jobs selected from job detail database **325**. Computer system **600** can be any computer capable of performing the functions described herein. (62) Computer system **600** can be any well-known computer capable of performing the functions described herein.

(63) Computer system **600** includes one or more processors (also called central processing units, or CPUs), such as a processor **604**. Processor **604** is connected to a communication infrastructure or bus **606**.

(64) One or more processors **604** may each be a graphics processing unit (GPU). In an embodiment, a GPU is a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

(65) Computer system **600** also includes user input/output device(s) **603**, such as monitors, keyboards, pointing devices, etc., that communicate with communication infrastructure **606** through user input/output interface(s) **602**.

(66) Computer system **600** also includes a main or primary memory **608**, such as random access memory (RAM). Main memory **608** may include one or more levels of cache. Main memory **608** has stored therein control logic (i.e., computer software) and/or data.

(67) Computer system **600** may also include one or more secondary storage devices or memory **610**. Secondary memory **610** may include, for example, a hard disk drive **612** and/or a removable storage device or drive **614**. Removable storage drive **614** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

(68) Removable storage drive **614** may interact with a removable storage unit **618**. Removable storage unit **618** includes a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **618** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **614** reads from and/or writes to removable storage unit **618** in a well-known manner.

(69) According to an exemplary embodiment, secondary memory **610** may include other means, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **600**. Such means, instrumentalities or other approaches may include, for example, a removable storage unit **622** and an interface **620**. Examples of the removable storage unit **622** and the interface **620** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

(70) Computer system **600** may further include a communication or network interface **624**. Communication interface **624** enables computer system **600** to communicate and interact with any combination of remote devices, remote networks, remote entities, etc. (individually and collectively referenced by reference number **628**). For example, communication interface **624** may allow computer system **600** to communicate with remote devices **628** over communications path **626**, which may be wired and/or wireless, and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **600** via communication path **626**.

(71) In an embodiment, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon is also referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **600**, main memory **608**, secondary memory **610**, and removable storage units **618** and **622**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **600**), causes such data processing devices to operate as described herein.

(72) Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. **9**. In particular, embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

(73) It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

(74) While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

(75) Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

(76) References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

(77) The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A computer-implemented method, comprising: retrieving, via a first service instance, an application programming interface (API) request from a client; storing, via the first service instance, an object representing the API request in a job detail database containing details related to the API request; determining, by a capacity computer module of the first service instance, an available bandwidth of the first service instance; transmitting, via the first service instance, the available bandwidth of the first service instance to a job processor; selecting, via the first service instance, a task associated with the object representing the API request stored in the job detail database based on the available bandwidth of the first service instance; in response to the selecting, locking, via the first service instance, the object stored in the job detail database, the locking preventing a selection of the object by a second service instance that is different from the first service instance; executing, via the first service instance, the selected task; and executing, via the second service instance, a second task associated with a second object representing a second API request stored in the job detail database, wherein the second service instance is different from the first service instance that received the API request; wherein at least one of the retrieving, storing, determining, transmitting, and selecting are performed by one or more computers.
2. The method of claim 1, wherein the executing the selected task comprises: executing, via the first service instance, the task based on templates that define standards for processing the task using interface definitions.
3. The method of claim 1, wherein the determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance comprises: determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including available memory.
4. The method of claim 1, wherein the determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance comprises: determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including available disk space.
5. The method of claim 1, wherein the determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance comprises: determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including an amount of memory required to process the API request.
6. The method of claim 1, further comprising wherein the determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance comprises: determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance at intervals.
7. A system, comprising: a memory; and at least one processor coupled to the memory and configured to: retrieve, via a first service instance, an application programming interface (API) request from a client; store, via the first service instance, an object representing the API request in a job detail database containing details related to the API request; determine, by a capacity computer module of the first service instance, an available bandwidth of the first service instance; transmit, via the first service instance, the available bandwidth of the first service instance to a job processor; select, via the first service instance, a task associated with the object representing the API request stored in the job detail database based on the available bandwidth of the first service instance; in response to the selecting, lock, via the first service instance, the object stored in the job detail database, the locking preventing a selection of the object by a second service instance that is different from the first service instance; execute, via the first service instance, the selected task; and execute, via the second service instance, a second task associated with a second object representing

a second API request stored in the job detail database, wherein the second service instance is different from the first service instance that received the API request.

8. The system of claim 7, wherein the at least one processor is further configured to: execute, via the first service instance, the task based on templates that define standards for processing the task using interface definitions.

9. The system of claim 7, wherein the at least one processor is further configured to: determine, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including available memory.

10. The system of claim 7, wherein the at least one processor is further configured to: determine, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including available disk space.

11. The system of claim 7, wherein the at least one processor is further configured to: determine, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including an amount of memory required to process the API request.

12. The system of claim 7, wherein the at least one processor is further configured to: determine, by the capacity computer module of the first service instance, the available bandwidth of the first service instance at intervals.

13. A non-transitory computer-readable medium having instructions stored thereon that, when executed by at least one computing device, cause the at least one computing device to perform operations comprising: retrieving, via a first service instance, an application programming interface (API) request from a client; storing, via the first service instance, an object representing the API request in a job detail database containing details related to the API request; determining, by a capacity computer module of the first service instance, an available bandwidth of the first service instance; transmitting, via the first service instance, the available bandwidth of the first service instance to a job processor; selecting, via the first service instance, a task associated with the object representing the API request stored in the job detail database based on the available bandwidth of the first service instance; in response to the selecting, locking, via the first service instance, the object stored in the job detail database, the locking preventing a selection of the object by a second service instance that is different from the first service instance; executing, via the first service instance, the selected task; and executing, via the second service instance, a second task associated with a second object representing a second API request stored in the job detail database, wherein the second service instance is different from the first service instance that received the API request.

14. The non-transitory computer-readable medium of claim 13, wherein the executing the selected task comprises: executing, via the first service instance, the task based on templates that define standards for processing the task using interface definitions.

15. The non-transitory computer-readable medium of claim 13, wherein the determining, by the capacity computer module of the first service instance, the available bandwidth comprises: determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including available memory.

16. The non-transitory computer-readable medium of claim 13, wherein the determining, by the capacity computer module of the first service instance, the available bandwidth comprises: determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including available disk space.

17. The non-transitory computer-readable medium of claim 13, wherein the determining, by the capacity computer module of the first service instance, the available bandwidth comprises: determining, by the capacity computer module of the first service instance, the available bandwidth of the first service instance based on parameters including an amount of memory required to process the API request.
